CSE STUDY HUB

Digital Forensics & Incident Response

UNIT 4 · Incident Response Management

What this unit is about

This unit builds understanding of **Incident Response Management**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Incident response purpose, policy, roles and preparation

2. Detection and identification: alerts, indicators and triage

3. Containment: short-term and long-term isolation

4. Eradication: remove malware, close entry points and reset credentials

5. Recovery: restore systems, monitor and validate services

6. Post-incident review, lessons learned and final report

7. NIST incident response lifecycle and organisational coordination

8. Evidence collection during response; preserve logs and chain of custody

9. Communication, escalation, legal notification and business continuity

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 4 NOTES · PART 1

Incident response purpose, policy, roles and preparation

A common incident-response cycle is preparation, detection/analysis, containment, eradication, recovery and lessons learned. Preserve evidence and document decisions while restoring services safely.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Detection and identification: alerts, indicators and triage

Detection and identification: alerts, indicators and triage is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Containment: short-term and long-term isolation

Containment: short-term and long-term isolation is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Incident Response Management, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 4 NOTES · PART 2

Eradication: remove malware, close entry points and reset credentials

Eradication: remove malware, close entry points and reset credentials is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Recovery: restore systems, monitor and validate services

Recovery: restore systems, monitor and validate services is a syllabus concept in Digital Forensics & Incident Response. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Post-incident review, lessons learned and final report

A common incident-response cycle is preparation, detection/analysis, containment, eradication, recovery and lessons learned. Preserve evidence and document decisions while restoring services safely.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Incident Response Management, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 4 NOTES · PART 3

NIST incident response lifecycle and organisational coordination

A common incident-response cycle is preparation, detection/analysis, containment, eradication, recovery and lessons learned. Preserve evidence and document decisions while restoring services safely.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Evidence collection during response; preserve logs and chain of custody

Chain of custody is the time-stamped record of who collected, handled, transferred, stored and examined evidence. Each action should identify the person, date/time, purpose and seal/identifier.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Communication, escalation, legal notification and business continuity

For Communication, escalation, legal notification and business continuity, discuss the objective, stakeholders and possible harm. Cover consent/authority, confidentiality, integrity, accountability, transparency and applicable policy or law. Recommend safeguards and explain how they are checked.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Incident Response Management, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Incident response purpose, policy, roles and preparation** — know the definition, core working and one example.

• **Detection and identification: alerts, indicators and triage** — know the definition, core working and one example.

• **Containment: short-term and long-term isolation** — know the definition, core working and one example.

• **Eradication: remove malware, close entry points and reset credentials** — know the definition, core working and one example.

• **Recovery: restore systems, monitor and validate services** — know the definition, core working and one example.

• **Post-incident review, lessons learned and final report** — know the definition, core working and one example.

• **NIST incident response lifecycle and organisational coordination** — know the definition, core working and one example.

• **Evidence collection during response; preserve logs and chain of custody** — know the definition, core working and one example.

• **Communication, escalation, legal notification and business continuity** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Incident Response Management. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Incident response purpose, policy, roles and preparation and explain its importance.

Q2. Explain the working/steps involved in Detection and identification: alerts, indicators and triage.

Q3. Differentiate between two related concepts from Incident Response Management.

Q4. Give one practical application of Communication, escalation, legal notification and business continuity and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Incident Response Management.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.